

Reducing Greenhouse Gases from Middle-Class Homes

by Don Fugler and Jane Thompson



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Could residents from this fairly sustainable neighborhood reduce their annual GHG emissions by an average of one metric ton?

Table 1. Participating Households

Attributes of Household	Number of Households	Percentage of Study Group
Number of Occupants		
1–2 persons	6	30%
3–4 persons	13	65%
5+ persons	1	5%
Year of Construction		
1900–1939	16	80%
1940–1979	3	15%
After 1980	1	5%
Environmental Awareness		
Poor	1	5%
Fair	11	55%
Good	8	40%

walking, or biking; garbage disposal; water usage; and other relevant household activities. This was followed by a questionnaire to link resource consumption with lifestyle patterns. The survey data and log information were used to generate a household report, which was given to the homeowners. This report showed the amount of GHG that their household emitted, broken down by sources and activities. It compared their resource consumption to that of their neighbors, and recommended specific actions that the homeowners could take to reduce their household's GHG emissions. The report included projected dollar savings that would result from adopting these measures.

To further help these homeowners, the study team coordinated bulk purchasing of environmental products,

In 1998 the Canadian federal government signed on to the Kyoto Protocol with a stated goal of reducing national greenhouse gas (GHG) emissions to 6% below 1990 levels by 2012. This is a difficult level to achieve, and every sector, including housing, was given targets and some incentives. Support for housing came in the form of rebates on energy-saving retrofits, which were available through the EnerGuide for Houses program, and consumer information programs. One such program, the One-Tonne (metric ton) Challenge, asked Canadian homeowners to reduce their annual GHG emissions from an average of 5 metric tons to an average of 4 metric tons—a 20% reduction—and suggested ways to achieve this reduction.

Jane Thompson, an architect who lives in one of Canada's earliest planned communities, on the outskirts of Ottawa, Ontario, decided to explore just how challenging it would be for her neighbors to meet this One-Tonne Challenge. In 2004 she received funding from the Canada Mortgage and Housing Corporation (CMHC) to undertake a neighborhood-based GHG reduction experiment. Could she get her neighbors to adopt environmentally sustainable behaviors by providing them with good information and the tools to help them achieve GHG reductions? Twenty households were selected from volunteers in her older neighborhood, which was developed in the 1920s (See Table 1). The neighborhood is quite sustainable already in that residents can walk to stores and services, and there

is ready access to public transport to nearby downtown Ottawa. By volunteering to participate in this year-long study, the occupants in these 20 households agreed to disclose their overall resource consumption at the beginning and end of the year; they also agreed to release their utility billing data for the one-year study period to the study team.

Involving the Homeowners

This neighborhood started with many advantages. The homeowners were generally well educated and environmentally conscious; for instance, 90% of the participating households already composted at the start of the study. Furthermore, the study group was composed of people who had volunteered for the project, making this group more likely to be environmentally concerned than the general population is. If this group could not achieve the desired savings, it seemed unlikely that Canadians in general would do better.

The study team supported the homeowners in many ways. First, it undertook an extensive survey of each house's characteristics and energy usage. An EnerGuide audit with a blower door test was performed on each house. The homeowners completed a household log for a week in October 2004, in which they noted all energy usage; trips by all modes of transportation—car, public transit,

such as rain barrels. Most of the households also participated in a pilot program to introduce “smart” electrical meters, which gave the homeowners and the study team better access to electrical consumption data. (The smart meter used was an Itron Centron model.) At the end of the monitoring year, gas and electrical consumption data were collected and were compared to gas and electrical consumption for the previous year. Vehicle mileage over the study year was recorded and compared to mileage during the one-week monitoring period in October. A community forum held in May 2005 and a postmonitoring questionnaire helped to assess the success of the measures taken.



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Twenty households from an older neighborhood in Ottawa, Ontario, volunteered to participate in a year-long GHG reduction study.

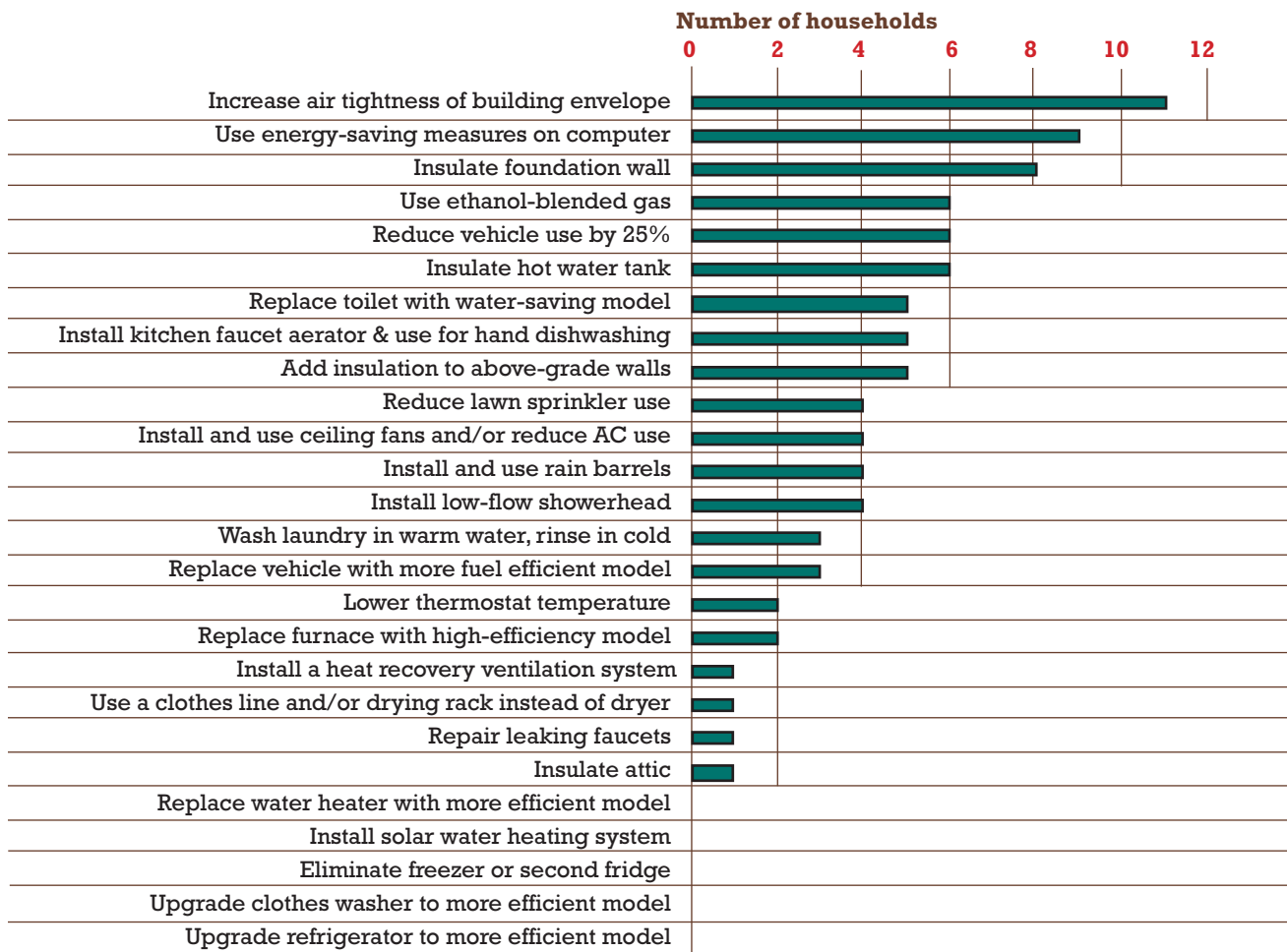


Figure 1. On average, homeowners adopted 26% of the recommended measures.

Table 2. Greenhouse Gas Reductions

House No.	No. of Occup.	GHG Per Household		GHG Per Occupant		
		Existing GHG	% of Recommended Measures Implemented	% Total Reduction in GHG	GHG at Start	GHG at End
1	4	12.2	17%	8%	3.1	2.8
2	2	11.7	0%	0%	5.9	5.9
3	3	19.5	0%	0%	9.8	9.8
4	4	29.9	14%	6%	7.5	7.1
5	4	15.9	21%	9%	4.0	3.6
6	n/a	n/a	n/a	n/a	n/a	n/a
7	2	13.7	24%	17%	6.9	5.7
8	4	27.5	73%	56%	6.9	3.0
9	4	13.9	16%	7%	3.5	3.2
10	3	9.1	18%	5%	3.0	2.9
11	5	13.5	38%	12%	2.7	2.4
12	4	14.9	14%	4%	3.7	3.6
13	3	13.9	18%	6%	4.6	4.4
14	1	16.7	26%	19%	16.7	13.6
15	4	9.8	39%	12%	2.5	2.2
16	4	14.0	11%	3%	3.5	3.4
17	2	7.6	64%	31%	3.8	2.6
18	4	10.9	31%	7%	2.7	2.5
19	n/a	n/a	n/a	n/a	n/a	n/a
20	2	19.3	43%	18%	9.7	8.0
Avgs	3.2	15.2	26%	12%	5.6	4.8

Varying Success

On average, homeowners adopted 26% of the recommended measures, which ranged from air sealing the home to insulating the attic (see Figure 1). Implementation of these measures resulted in projected average household GHG reductions of 12% (see Table 2). Per person GHG reduction averaged about 0.8 metric tons, or just short of the One-Tonne Challenge.

Household GHG reductions ranged from 0% to 56%. Two households did not provide follow-up data after one year. One of these households moved during the year and one objected to the comparisons being made and withdrew from the study, although this household's utility records were still available and surprisingly showed some of the largest reductions in the group.

Utility bills for the year before and the year of the study were compared, to see if measurable energy savings had been achieved. Because households implemented energy-saving measures throughout the year of the study, energy savings for the years following that year should be higher than savings for the year of the study itself. Overall, natural gas consumption dropped by

9% during the year of the study. This was a year when there were 1% fewer heating degree-days than there were in the previous year. Mean electricity consumption actually rose by 9%. This was partly because the summer was warmer than usual, and homeowners made more use of air conditioning. Water consumption dropped by 14% over the previous year. However, the effect of this drop on GHG emissions is hard to quantify.

Household 8 achieved the greatest GHG reduction, at 56%, or over 15 metric tons. They did this by implementing 12 of the 16 recommendations made to them. The most significant reductions related to their automobile use. They replaced two cars, one of which was very fuel inefficient, with more efficient automobiles. They reduced the amount they drove by 25%, and they switched to ethanol gasoline. Measures that they took to increase the energy efficiency of their house included adding insulation; increasing airtightness; installing low-flow fixtures, rain barrels, and ceiling fans; and reducing their use of air conditioning. Household 17 is also

noteworthy. This household began the study with the lowest GHG emissions in the group at 7.6 metric tons total, or 3.8 metric tons per occupant, and still found ways to reduce their total emissions by 30% or 2.4 metric tons.

The differences in environmental impact per occupant were striking (see Figure 2). Home heating fuel, electricity, and water consumption per occupant was 5.5 to 7.5 times greater in the highest-consuming household than it was in the lowest-consuming household, and the average consumption was about 2.5 times the lowest consumption. Vehicle emissions and waste production were even more variable, with highest levels at 11 to 21 times the lowest levels, and average levels at about 6 times the lowest levels. Total GHG emissions per occupant were 5.5 times greater in the household with the highest environmental impact than they were in the household with the lowest environmental impact. These variations indicated that, at least for this population group, it is not accurate to assume that individual Canadians are producing GHG emissions at roughly similar rates, nor is it realistic to ask them to cut emissions by the same amount.

Large Homes, Larger Emissions

Who was using more, on a per capita basis? People who lived in relatively large houses or who just had more space per occupant—particularly those with more than 100 square meters (1,000 square feet) of floor area per occupant. (The four largest houses ranged in size from 2,300 to 2,600 square feet. Sadly, this is about the same size as the average new house in the United States.) These families typically consumed more heating fuel, electricity, and water. This could be because bigger houses tend to consume more. Or it could be a function of lifestyle, or the simple mathematical result of dividing household GHG production by a smaller number of occupants. Most of the families with the highest GHG emissions per occupant consisted

of two adults, both over 40 years of age. Higher than average automobile emissions were also a big factor in each of the six top GHG-emitting households. Not surprisingly, two-car families almost invariably had higher mileage than single-car families. These lifestyle factors appeared to have more of an impact on GHG emissions than the energy-saving features of a house that we think of most often, such as the tightness of the building envelope or the efficiency of the appliances and fixtures.

The four households in the study group with the lowest GHG emissions lived in homes that were modest in size for the number of occupants. Their homes had been carefully renovated to increase insulation values, airtightness, and the energy efficiency of appliances and fixtures. Their car use was low, or they owned very efficient vehicles. Finally, these households were consciously adopting conservation practices. Although these four households were already performing better than the other households in the study, all four elected to implement 30% to 40% of the measures recommended to them. This brought their GHG emissions down from an already-low rate to between 2 and 2.5 metric tons per person.

Participants were surveyed at the start of the project on their knowledge of energy-saving options and on what they personally did to save energy, as compared to their neighbors. There was little correlation between these self-assessments and actual GHG generation rates.

Implications for GHG Reduction Programs

This was not your typical weatherization project. Compared to participants in most U.S. weatherization programs, participants in this study group were more environmentally aware, had more discretionary income, were very willing to participate in the study, and benefited from having a helpful, knowledgeable group of



A majority of the study participants concluded that the most effective way to reduce GHG emissions would be to mandate stricter environmental standards.

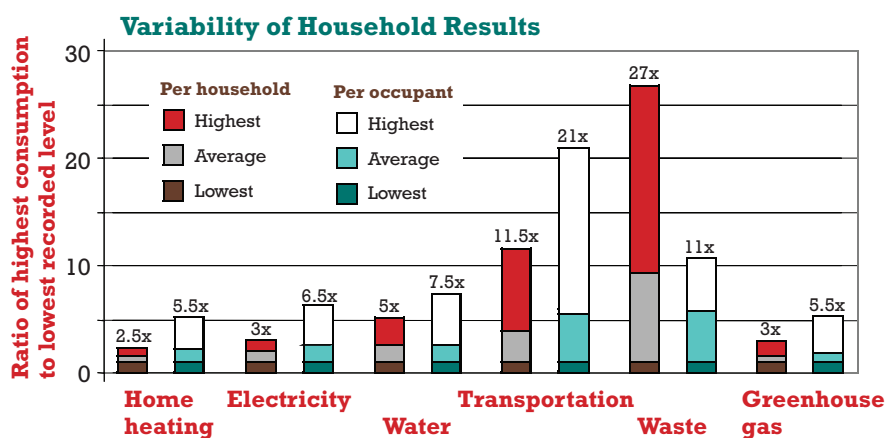


Figure 2. The difference in reduction of GHGs by occupant are striking.

consultants to advise them. Despite all these advantages, savings of heating fuel and electricity were marginal, and GHG reductions were still hard to achieve, especially on a group basis. The bright spots were that significant reductions could be achieved, and that participants who followed through on the advice they were given will be paying less for energy and contributing less to GHG production. The most dispiriting result is that, even with all the factors lined up to promote the success of this GHG saving effort,

many of the participants made few or no changes.

Participants were asked to list their priorities in making decisions about home upgrades. Their top priorities were improving comfort and lowering operating costs. Reducing environmental impact and improving air quality ranked as lower priorities. The major obstacles to reducing environmental impact were seen as financial cost and lack of time and knowledge to evaluate and implement environmental measures. In keeping with this emphasis on practical,

cost-effective measures, the majority of the environmental measures that were implemented were those that the household report identified as high priority, with a payback period of less than ten years. Measures with a payback period of more than ten years tended not to be adopted in the first year. These included more expensive items, such as window replacement, solar water- heating systems, and upgraded appliances.

Participants were asked to evaluate the environmental monitoring technique used in the study. They found it more useful to them than existing government-sponsored projects, such as the One-Tonne Challenge, but most said that relying on voluntary programs would not work. Instead, a majority of participants stated that the most effective way for the government to reduce GHG emissions would be to legislate higher mandatory environmental standards for house construction, car fuel efficiency, appliances, equipment, packaging—and whatever else a consumer consumes.



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For more information:

To find out more about the Kyoto Protocol, visit http://unfccc.int/kyoto_protocol/items/2830.php.

To find out more about the Itron Centron meter that was used in this study, visit www.itron.com.